NAME: Gaurika Mittal

BATCH: 15

SAP ID: 590034892

EXPERIMENT-4

Question

The following problem is to find the product of two integers without using the multiplication operator *. Your teacher has given an incorrect algorithm and an incorrect flowchart for this problem. Study both carefully and complete the tasks given below.

Incorrect Algorithm

Algorithm: Multiply Two Numbers Without Using * Operator

Step 1: Define num1**,** num2**,** prod**,** index as integers.

Step 2: Input the values of num1 and num2.

Step 3: Set prod = 0 and index = 1.

Step 4: Repeat the following steps until index <= num2.

- **Step 4.1:** `prod = prod + num1`
- **Step 4.2:** `index = index + 1`

Step 5: Display prod.

Step 6: Stop.

1. Mistakes in the Given Algorithm

1. The algorithm does not check whether either input number is **0**. If one of the numbers is zero, the product should be zero immediately.

2. It assumes that num2 is always positive. If num2 is negative, the loop condition index <= num2 becomes false at the beginning, so the loop never executes.
3. It does not handle negative numbers. Therefore, it gives incorrect results for:
    - positive × negative
    - negative × positive
    - negative × negative
4. It always adds num1 exactly num2 times. It does not use the smaller absolute value, so it performs unnecessary additions.
5. The algorithm does not determine whether the final answer should be positive or negative.
6. The algorithm does not satisfy the requirement of using the minimum possible number of additions.

2. Mistakes in the Given Flowchart

1. The flowchart does not check if either input number is zero.
2. It does not convert negative numbers into positive numbers before starting the loop.
3. The decision box checks only index <= num2, which works only when num2 is positive.
4. There is no step to determine the sign of the final product.
5. The flowchart always repeats addition num2 times instead of using the smaller absolute value.
6. It cannot correctly handle negative inputs.

3. Corrected Algorithm

Step 1: Start.

Step 2: Declare num1**,** num2**,** product**,** count**,** value**,** sign as integers.

Step 3: Read the values of num1 and num2.

Step 4: If either number is 0, display Product = 0 and stop.

Step 5: Set sign = 1.

Step 6: If num1 is negative, make it positive and change the value of sign.

Step 7: If num2 is negative, make it positive and again change the value of sign.

Step 8: Compare the two numbers. Store the smaller value in count and the larger value in value.

Step 9: Set product = 0.

Step 10: Repeat until count becomes 0.

- Add value to product.

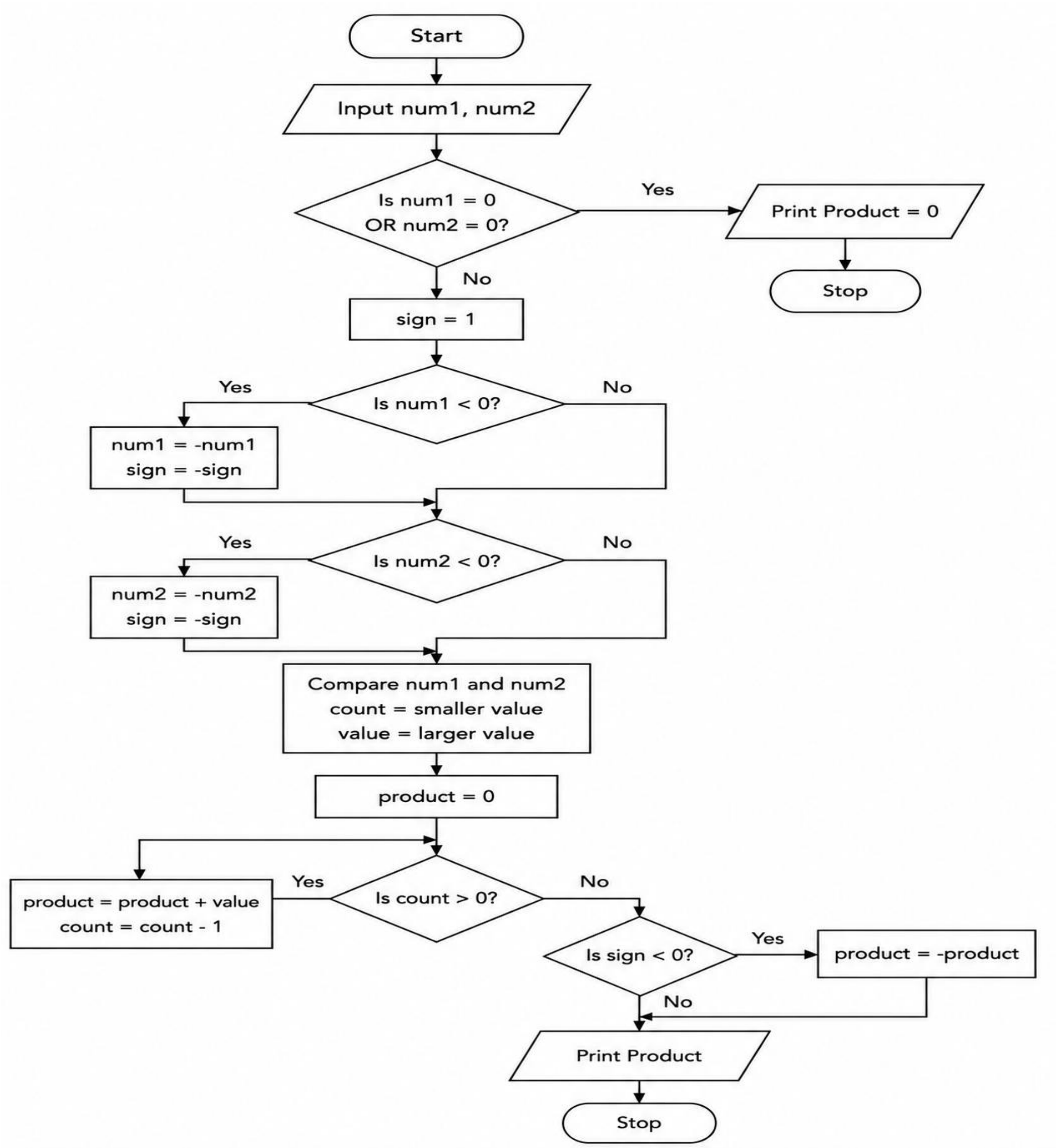
- Decrease count by 1.

Step 11: If sign is negative, change the product to a negative value.

Step 12: Display the product.

Step 13: Stop.

4. Corrected Flowchart



5. Verification

```c
#include <stdio.h>
int main()
{
    int num1, num2;
    int product = 0;
    int sign = 1;
    int count, value;
    printf("Enter two integers: ");
    scanf("%d %d", &num1, &num2);
    if (num1 == 0 || num2 == 0)
    {
        printf("Product = 0");
        return 0;
    }
    if (num1 < 0)
    {
        num1 = -num1;
        sign = -sign;
    }
    if (num2 < 0)
    {
        num2 = -num2;
        sign = -sign;
    }
    if (num1 < num2)
    {
        count = num1;
        value = num2;
    }
    else
    {
        count = num2;
        value = num1;
    }
    product = 0;
    while (count > 0)
    {
        product = product + value;
        count = count - 1;
    }
    if (sign < 0)
    {
        product = -product;
    }
    printf("Product = %d", product);
    return 0;
}
```

Using this program I have checked every test cases below and they are running very nicely.

Test Case	Input	Expected Output	Minimum Additions
1	12 3	Product: 36	3
2	3 12	Product: 36	3
3	-6 5	Product: -30	5
4	6 -5	Product: -30	5
5	-9 -4	Product: 36	4
6	-4 -9	Product: 36	4
7	0 8	Product: 0	0
8	8 0	Product: 0	0
9	0 -7	Product: 0	0
10	-7 0	Product: 0	0
11	0 0	Product: 0	0
12	1 -7	Product: -7	1
13	-1 -9	Product: 9	1

Algorithm picture is generated by Chatgpt